

# Generalizability of Remote Sensing Based Data-Driven Plot-Scale in-Season Crop Yield Forecasting Models

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## Introduction

- Accurate plot scale crop yield are essential for identifying spatial variability, determining key factors influencing yield variability, and providing site-specific management insights.
- Further, such yield forecasts are crucial for deciding grain storage and market planning.
- Remote sensing and plant phenotyping data along with data-driven models have the potential to provide reliable forecasts.
- Most existing data-driven models provide yield forecasts at county or field scales.
- Limited models exist for in-season, plot scale spatial yield predictions, but lack in-season forecasting capabilities.

## Objectives

- To develop and compare plot-scale, in-season yield forecasting models for corn and soybean using:
  - Light Gradient Boosting Machine (LightGBM),
  - Attention-based Long Short-Term Memory (ALSTM),
  - Encoder-only Transformer (TF) architectures.
- To evaluate the generalizability of these models using:
  - independent plot-level data, and
  - an unseen growing year.

## Study area

- Field Plant Phenotyping Facility, Eastern Nebraska Research, Extension, and Education Center, Ithaca, NE (41°08'44.1" N, 96°26'20.5" W)
- Plot size: 6.1 m x 4.6 m
- Crop yield data was collected from the two central rows

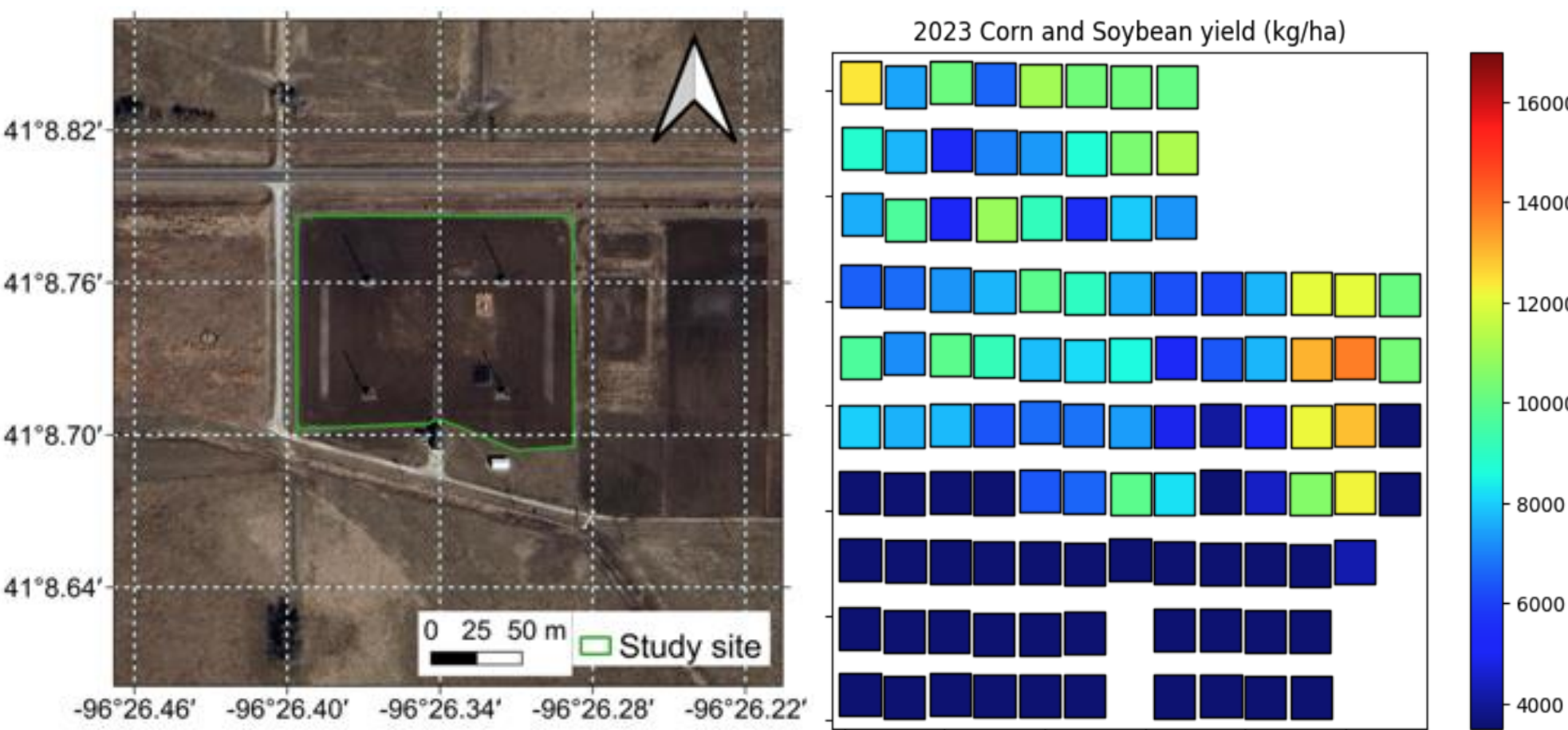


Fig. 1. Location of the experimental field phenotyping site equipped with NU Spidercam System (left) and crop yield (corn and soybeans) for each plots in the 2023 season (right)

## Data

- Sensors in NU Spidercam system: Multi-spectral, thermal infra-red sensors, spectrometer and 3D LiDAR.
- Data were collected twice a week during the growing season.

Table 1. Details of input data

| Data                                      | Description                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------|
| Vegetation Indices (VI): 13 count         | NDVI, OSAVI, GNDVI, GRVI, NDRE, NDREI, GCI, RECI, SCCCI, VARI, SIPI, SI, RVI |
| Solar Induced Fluorescence (SIF): 9 count | Ratio between narrow spectral bands. Indicator of photosynthetic activity.   |
| Phenotypes: 3 count                       | Canopy height, Plant coverage, canopy average temperature                    |
| Soil average temperature                  |                                                                              |

## Methods

- 13 models each of LightGBM, ALSTM and TF-based plot scale in-season yield forecasting models were trained separately for corn and soybean crop following protocols in Fig. 2.
- A total of 78 models were trained:
  - Corn crop: 3 models x 13 time-steps = 39 count
  - Soybean crop: 3 models x 13 time-steps = 39 count

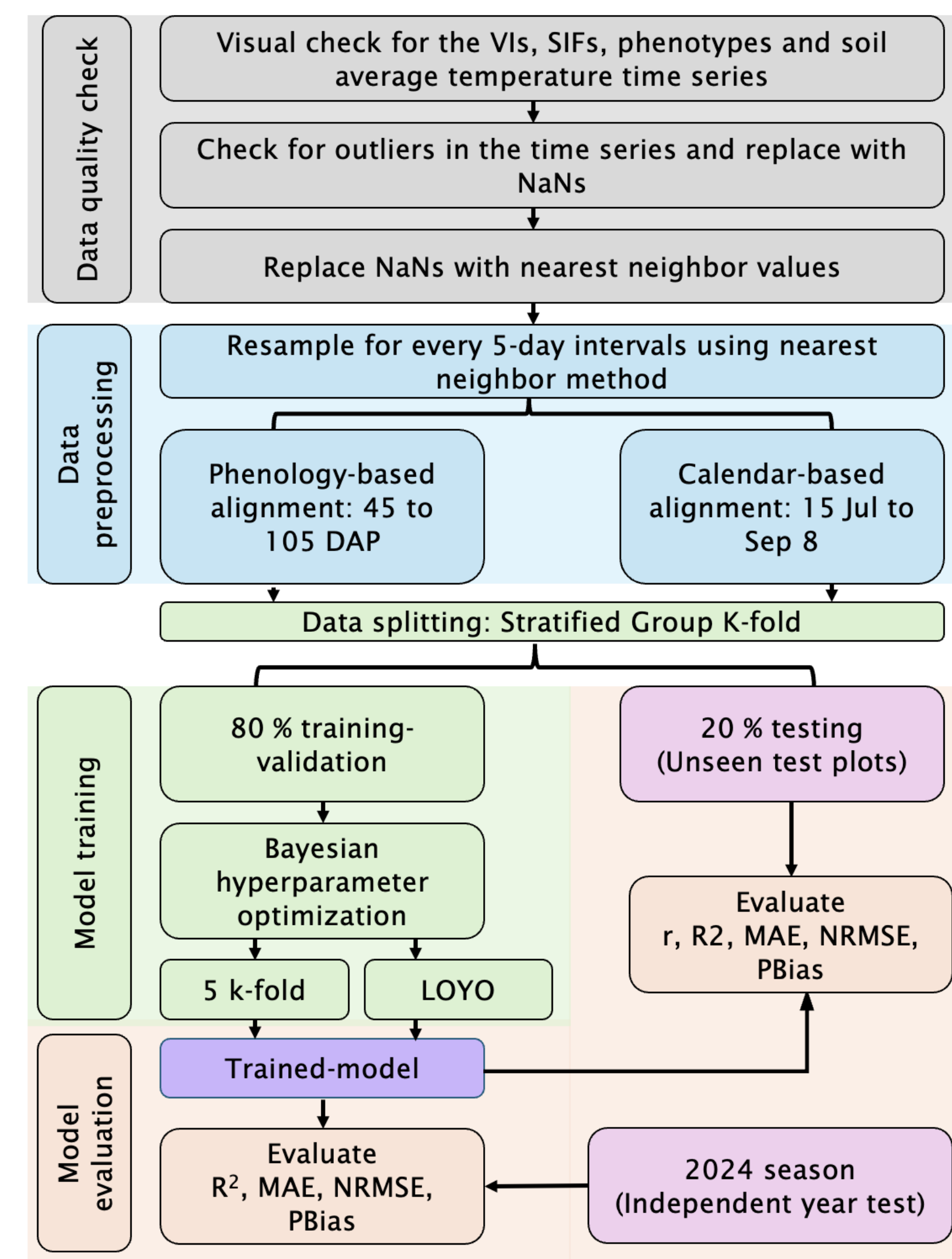


Fig. 2. Schematic flowchart illustrating model training and evaluation protocols

## Results

- Generally, the accuracy increased with increasing time series data points during the growing season (Fig. 3).
- For unseen test plots, the mean Coefficient of determination ( $R^2$ ) across all DAP for LGBM, ALSTM and TF models were 0.839, 0.675 and 0.768 respectively for corn. The same for soybean were 0.834, 0.774 and 0.848 (TF), respectively.

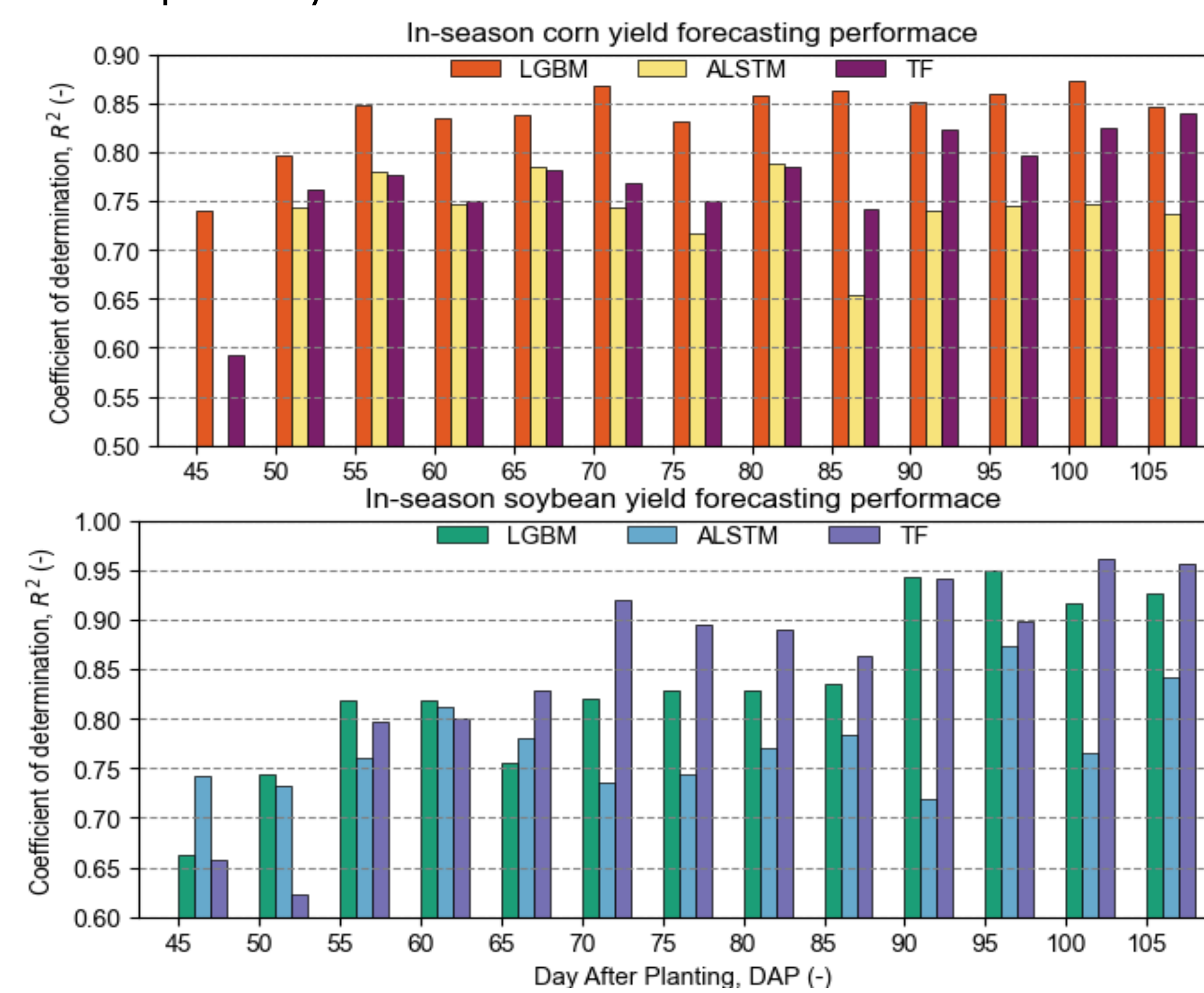


Fig. 3. Accuracy for in-season corn (top) and soybean (bottom) crop yield forecasting models in unseen test plots

- All the models failed to forecast yield in the next season as indicated by negative or low  $R^2$  values (Fig. 4).
- Comparatively, the LGBM performed best for corn and ALSTM for soybean yield forecasting in independent year.

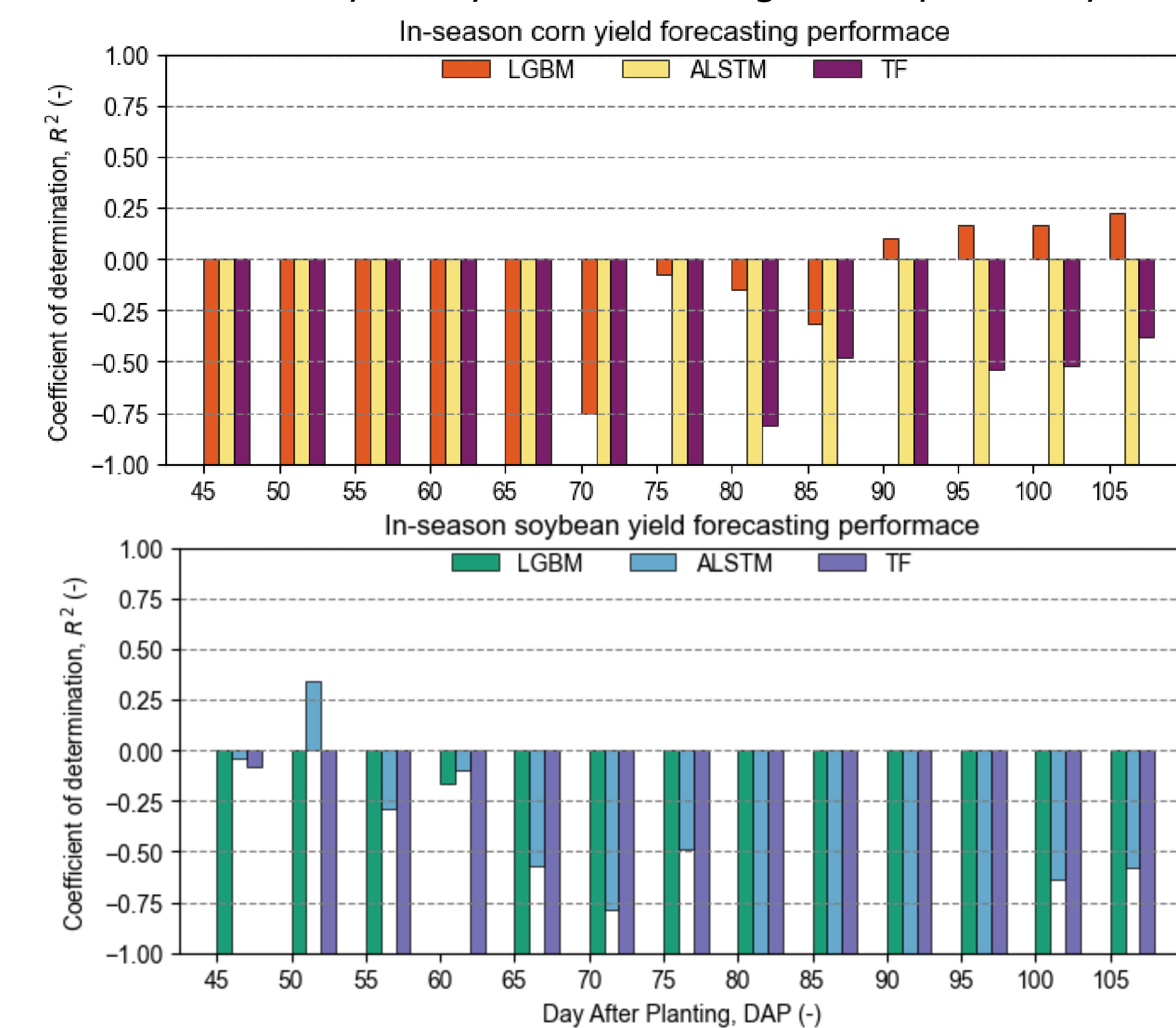


Fig. 4. Accuracy for in-season corn (top) and soybean (bottom) crop yield forecasting models in independent year

## Discussions

- As the models performed accurately in unseen test plots, but not in independent year, practical implementations of the models would require some prior yield forecasts data for that season. These prior yield forecast could potentially be obtained from county-level yield forecasting systems. However, it will require further fine tuning to accommodate for the discrepancies in spatial scale.
- ALSTM and TF models are constrained by the limited crop yield samples, leading to comparatively lower performance than LGBM.
- Additionally, yield forecasting is a time series feature extraction problem in addition to time series regression problem. Therefore, using the ALSTM or TF model architectures to systematically extract features at each critical growth stages rather than the entire growing season may aid in developing reliable forecast.
- The inability for the model to predict in independent year is likely due lack of temporal yield samples i.e., only 5 years. This limits the ability of the model to capture climate variability (>10 years).
- Further, the distribution of the crop yield data in the training year and testing year may also have contributed to the low performance.

## Conclusions

- LGBM, ALSTM and TF models can forecast corn and soybean yield in unseen test plots with mean  $R^2$  higher than 0.675.
- Performance of the model gets better with increasing DAP, indicating that mid to late season forecast are more reliable than early season forecast.
- All the models did not generalize on independent year. This can be attributed to lack of temporal yield data samples for training. Without sufficient year-to-year samples, the models failed to learn patterns that generalize across different growing seasons.

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